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(54) **VALVE BLOCK, TOOL ATTACHMENT, A WORKING MACHINE AND THE USE OF A VALVE BLOCK**

VENTILBLOCK, WERKZEUGANSATZ, ARBEITSMASCHINE UND VERWENDUNG EINES
VENTILBLOCKS

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Description

Field of the invention

[0001] According to a first aspect the present invention relates to a valve block for use with a tool attachment of a working machine, where the tool attachment is adapted for connection to different types of hydraulically manoeuvred tools or implements with the aid of a hydraulically actuated locking device.

[0002] According to second, third and fourth aspects the invention relates to a tool attachment, a working machine and to the use of a valve block according to the invention.

Description of the background of the invention

[0003] During the last ten to fifteen years the use of tool attachment systems for the rapid change of buckets and tools with regard to excavating machines, wheel-mounted loaders, material handling machines of several different types have been a matter of course in the majority of European markets. The development of such systems has quickly advanced from simple mechanical solutions in which the driver leaves the driver's cabin and releases or locks the tool attachment mechanism with the aid of a simple movement of a lever and manually connects the working hydraulics of the tool. This solution has gradually been replaced with hydraulic solutions in which the driver is able to control a locking cylinder that actuates the attachment locking mechanism, by simply pressing a button in the driver's cabin. Although the tool is quickly connected mechanically in this latter case, it is still necessary to connect the hydraulic hoses to the hydraulic tool by hand. In recent years there have also been used fully automatic tool fastening systems. One example of such systems is described for instance in EP 483 232 and EP 602 165. In these cases mechanical and hydraulic buckets and tools are coupled and released fully automatically directly from the driver's cabin. Hydraulic hoses and electric supply lines are also coupled fully automatically in these cases.

[0004] The solutions described above apply generally to all types of tool carriers. For the sake of simplicity, however, the following text will be concentrated on the tool attachments for wheel-carried loaders.

[0005] The system solutions relating to fully automatic tool attachment solutions have become more complex and a number of sub-functions have been proposed in an endeavour to achieve fully functional solutions, namely:

a) A hose fracture function which is normally fitted directly to the locking cylinder to ensure that the locking function of the attachment will function even in the event of a hose fracture on the hydraulic hoses leading to the locking cylinder.

b) An electricity reversing function. In conjunction with the hydraulic rapid tool attachment the most usual solution applied involves the installation of an electricity reversing valve, a type 6/2 valve, in order to lend oil from the normal tool hydraulic system to the locking function of the attachment locking pistons. This magnetic valve is controlled electrically via a low voltage cable and a contact breaker in the driver's cabin. The reversing valve is normally placed on the machine, on its arm or jib, or separately on the tool attachment. By placing the valve as close as possible to the locking cylinder, it is possible to reduce the length of the hydraulic hose for the locking function. The valve is usually placed on the tool attachment, although this incurs serious mounting problems, especially in the case of smaller machines, due to the highly confined space available to this end. The extension of the hose between the valve and the locking cylinder presents another problem.

c) Post tensioning of the locking function in the above solution via the reversing valve. When the locking hydraulics is released with the aid of an el-type reversing valve the locking cylinder will not be subjected to any post-tensioning pressure when the valve is short circuited to the other function to give working hydraulics ?. The post tension on the locking system can thus be solved with a separate arrangement, in which a so-called bypass line that includes a check valve is drawn between tool hydraulics and the working hydraulics. See the following hydraulic diagram. This is usually solved by placing a hose loosely on the tool attachment. WO 2004/067855 describes a valve solution in respect of the quick attachment of a kran?, where said post-tensioning function is included in a separate valve block. It will be noted that this block is not connected directly to the cylinder but is, instead, connected via hydraulic hoses.

d) The load is relieved automatically prior to connecting and disconnecting the tool hydraulics. This enables the pressure in the hoses leading to the tool hydraulics to be eliminated so as to enable the hydraulics to be released in the absence of pressure. This is particularly important in connection with fully automatic attachment systems, so that the seals in the quick couplings will be protected. This automatic pressure relief is normally effected on the machine by installing leakage lines from the tool hydraulics that are connected directly to a tank via a dumping valve. The dumping valve is controlled automatically from the attachment locking hydraulics and is provided with a time relay which ensures that the tool hydraulics are relieved of pressure prior to connection and disconnection.

[0006] All of these sub-functions are normally installed

separately on the machine, the excavating/lifting arm? or on the tool attachment. This installation is normally carried out in the aftermarket and is expensive and complicated since it is often necessary to loosen these parts individually from case to case and since these parts are tailored separately for the machine concerned. Moreover, several parts are included in the various solutions, in the form of machine suppliers, tool suppliers or attachment suppliers. In the maturity of cases this will result in an obscure distribution of responsibility between the parties concerned. The object of the present invention is to avoid the problems that are associated with conventional arrangements of these functions.

Disclosure of the invention

[0007] The objects of the invention mentioned above are achieved with a valve block according to claim 1. Those drawbacks associated with the traditional design of said means are eliminated by the inventive valve block due to most of these functions being included in a single unit which may, of course, be delivered as a part of the tool attachment. The tool attachment may thus be delivered with a number of these functions already embodied thereby obviating the need for subsequent supplementation or requiring only at least less supplementation. All that is required in respect of mounting the tool attachment to the machine are two hydraulic lines and an electric cable connected to the machine.

[0008] According to a preferred embodiment of the inventive valve block, the valve members are adapted to fulfil at least one of said functions, preferably both of the functions.

[0009] The benefits mentioned above are thus further accentuated due to the fact that the requirement for separate solutions is further reduced. An optimal design, primarily in the case of sophisticated systems, is, of course, achieved when all of the functions are integrated in the valve block, therewith completely eliminating the requirement of a separate solution for each function.

According to another preferred embodiment, the valve block includes attachment means for attachment of the valve block to the locking device.

This enables the valve block to be readily fastened to the tool attachment, which in turn facilitates mounting of these components in a well defined fashion. Because the valve block is fastened to the locking device of the tool attachment, the need to arrange lines between valve block and tool attachment is eliminated.

[0010] According to the invention, the valve members include a first check valve which is adapted to provide the first function. The check valve ensures in a simple manner that the hydraulic connection between the locking device and the hydraulic system of the working machine will be cut off simultaneously when the pressure in the hydraulic system decreases drastically, as a result of a fractured hose or the like for instance.

According to another preferred embodiment of the inven-

tion, the valve members include a reversing valve which is adapted to provide the function which enables said first port means to be connected selectively to either said second port means or to said third port means.

5 This function is achieved fully satisfactorily in an optimal manner with regard to simplicity and reliability with the aid of a reversing valve.

According to another preferred embodiment, the reversing valve is operated electro magnetically. This facilitates positive operation of the reversing valve from the driving cabin of the working machine and reduces the need of changing the mode of the valve manually.

10 According to another preferred embodiment of the invention, the line or conduit system in the valve block comprises at least one line which includes a check valve and which connects together the second and third port means.

[0011] According to the second and third aspects of the invention, the object of the present invention is also achieved by providing a tool attachment with a valve block according to the present invention or with one of the preferred embodiments of the invention and by providing a working machine with such a tool attachment respectively.

20 According to the fourth aspect of the invention, the object of the invention is achieved by using a valve block for connecting and manoeuvring a working tool.

[0012] As a result of the inventive tool attachment, the working machine and the use respectively afford benefits of a corresponding sort with respect to the inventive valve block and the preferred embodiments described above.

[0013] The invention will now be explained more specifically with reference to the following description of a beneficial embodiment thereof and also with reference to the figures of the accompanying drawings.

Brief description of the figures

[0014]

40 Figure 1 is a perspective view of a valve block according to the invention and a tool attachment locking device.

45 Figure 2 illustrates a hydraulic circuit for the valve block and the components connected thereto.

Description of beneficial embodiments of the invention

50 **[0015]** Figure 1 is a perspective view of a valve block 1 according to the present invention. The valve block 1 is adapted so that it can be screwed firmly to a tool attachment locking device 7. The locking device is comprised of a hydraulic cylinder 7 onto which a connecting block 8 is firmly welded. The valve block 1 is intended to be screwed firmly to the connecting block of the hydraulic cylinder by means of a screw 23. A seal 28 is provided between the valve block 1 and the connecting block 8.

The hydraulic cylinder 7 is mounted on a tool attachment (not shown) of a working machine and includes a piston rod 30, 31 at each end. The piston rods 30, 31 can be manoeuvred between an outer and an inner position. In the outer position respective piston rods 30, 31 co-act with a locking mechanism of the working tool or implement (not shown) which is connected to the tool attachment such that the working tool will be firmly connected to the tool attachment? In the inner position of the piston rods the locking action is released so as to enable the tool to be released from the tool attachment. This type of locking mechanism is known generally and has various variants. Any closer description of this locking mechanism would therefore be superfluous in this connection.

[0016] The valve block 1 includes port means for the connection of hydraulic lines. First port means, which include the ports 17, 18, are intended for connection to hydraulic hoses on the working machine and are manoeuvred from the driving cabin of the working machine. Other port means, which include the ports 19, 20, are intended for connection to hoses that are joined to female quick-couplings on the tool attachment. Third port means, which include two ports 21, 22 (obstructed in Figure 1) are connected to the hydraulic cylinder 7 via the connecting block. The port means communicate with one another through a line system in the valve block 1, which will be described more specifically with reference to Figure 2. There is mounted in the valve block 1 a unit which includes an electro magnet 3 for reasons described below. The electro magnet 3 is provided with a connection 24 for a cable that leads to the driving cabin of the working machine.

[0017] Figure 2 illustrates the line system within the valve block 1 and adjacent equipment. The port 17, 18 of the first port means of the valve block are connected to hydraulic hoses 32, 33 on the working machine, these hydraulic hoses being manoeuvred from the driving cabin of the working machine. The ports 19, 20 of the second port means of the valve block are each connected to a female quick-coupling 10 disposed at a displaceable coupling arm 11, wherewith the female quick-couplings 10 can be coupled to male quick-couplings 12 in a coupling block 13 on the tool attachment. The male quick-couplings 12 are connected to the hydraulic cylinder 15 of the tool attachment via hydraulic hoses 14. The ports 21, 22 of the third port means of the valve block are connected to the locking cylinder 7 via the connecting block 8.

[0018] The valve block 1 includes a system of lines and valves. A first pair of lines 34, 35 connects the ports 17, 18 with the reversing valve 2. A second pair of lines 36, 37 connects the ports 19, 20 with the reversing valve 2, and a third pair of lines 38, 39 connect the ports 21, 22 with the reversing valve 2.

[0019] The line 38 includes a pilot-controlled check valve 4 which is intended to prevent flow from the port 21 to the port 17. Disposed between the line 36 and the line 38 is a connecting line 40 that includes a check valve 6 which is intended to prevent flow from the line 38 to the

line 36. A corresponding connecting line 41 that includes a check valve 6 is located between the line 37 and the line 38. Disposed between the line 38 and the line 39 is a connecting line 42 that includes a pressure limiting valve 5. The reversing valve 2 is shown in the figure in a position in which the ports 17, 18 of the first port means are connected with the ports 19, 20 of the second port means, which is the case when the hydraulics are used for manoeuvring of the tool. The reversing valve 2 can be moved to the left in the figure by means of the electro magnet 3, to a position in which the ports 18, 17 of the first port means are connected with the ports 21, 22 of the third port means, which is the case when the locking cylinder 7 shall be manoeuvred for locking or for releasing the tool. The reversing valve 2 thus enables the hydraulics of the working machine to be used either for releasing the tool or for connecting the tool.

[0020] When a switch shall be made from one tool to another, the reversing valve is manoeuvred to the position in which the ports 17, 18 are connected with the ports 21, 22. The connection between the hydraulics of the working machine and the hydraulic cylinder 15 of the tool is therewith broken and the connection is made instead to the locking cylinder 7. The line 39 is therewith pressurized, causing the piston rods 30, 31 to move inwards via the lines 26, 27 so as to release the tool from the locking position. The quick couplings 10, 12 are, at the same time, separated from each other owing to the fact that the locking cylinder 7 also controls mechanically the engagement and disengagement of the quick couplings. A fresh working tool is then applied, wherewith the line 38 is pressurized and therewith urges the piston rods outwards to a locking position. At the same time, female quick-couplings 10 are coupled to the male quick-couplings on the fresh tool. The reversing valve 2 is then moved back to the position shown in Figure 2, for manoeuvring the fresh tool. The line 38 is kept pressurized so as to ensure that the locking state remains.

[0021] The check valve 4 in the line 38 ensures that the pressure in the line 38 will be sustained, and therewith also the locking effect, even in the event of a hose rupture or a hose malfunction.

[0022] The connecting lines 40, 41 and their check valves 6 are intended to post-tension the locking function on the one hand and to relieve the working hydraulics of load prior to engaging and disengaging the working tool. When the reversing valve is in its working hydraulic mode, as illustrated in Figure 2, the pilot-controlled check valve 4 is blocked. If the pressure in the working hydraulics exceeds the pressure in the line 38 to the locking side of the locking cylinder oil will be subsequently delivered via the check valves 6. Because the system has such a check-valve connection to each of the lines 36, 37 to the working hydraulics, after-supply will take place from the working pressure line 36, 37 that has the highest pressure. This ensures that the tool and the hydraulic couplings 11, 12 will always be in a fully coupled position.

[0023] When the reversing valve 2 is set to a position

for a tool change, i.e. moved to the left in Figure 2 would be ports 21, 22 thus connected to the hydraulics of the working machine, any remaining pressure in the hydraulic line 36, 37 will be drained to a tank via the check valves 40, 41. This being due to the fact that the pressure in the lines 36, 37 is higher than the pressure in the line 38 which is then pressure free up to the tank.

[0024] The purpose of the pressure limiting valve 5 is to protect the sealing system in the locking cylinder 7 and to enable manoeuvring of the reversing valve 2.

Claims

1. A valve block (1) with a line system for use in respect of a tool attachment of a working machine, where the tool attachment is adapted for connection to different kinds of hydraulically operated tools or implements with the aid of a hydraulically actuated locking device (7) that the valve block (1) includes first port means (17, 18) for connection to hydraulic lines (32, 33) of the working machine, second port means (19, 20) for connection to hydraulic coupling means (10) adapted for coupling to hydraulic coupling elements (12) of the tool or implement, and third port means (21, 22) for connection to the locking device (7), that a valve means (2) is provided in the valve block (1) with a function which enables said first port means (17, 18) to be connected selectively to either said second port means (19, 20) or to said third port means (21, 22), that the line system includes at least one connecting line (40, 41) which is provided with a valve means in the form of a check valve (6) and which connects said second port means (19, 20) and said third port means (21, 22) so as to obtain a post-tensioning of the locking device (7) when the first port means (17, 18) are connected to said second port means (19, 20) and for relieving load prior to connecting and disconnecting the hydraulic connection (12) of the working tool; **characterized in that** the line system in the valve block (1) communicate with said port means and also include valve means (4, 5), wherein said valve means (4, 5) are adapted to fulfil at least one of the following functions:

- a first function for ensuring the presence of hydraulic pressure in the locking device (7) in the event of a fracture or a leakage in the hydraulic lines (32, 33) of the working machine by a first check valve (4) which is intended to provide said first function; and
- a second function for achieving pressure limitation of the locking device (7) in that the third port means (21, 22) include two ports (21, 22) which are each connected to a hydraulic line (38, 39), and **in that** between which hydraulic lines there is provided a connecting line (42) which includes a pressure limiting valve (5) and which

is adapted to provide said second function.

2. A valve block (1) according to claim 1, **characterized in that** said valve means (4, 5) are adapted to fulfil all of said functions.
3. A valve block (1) according to any one of claim 1 - 2, **characterized in that** the valve block (1) includes means (23) for fastening the valve block (1) to the locking device (7).
4. A valve block according to any one of claims 1 - 3, **characterized in that** said valve means (2, 4, 5, 6) include a reversing valve (2) which is adapted to provide said function which enables said first port means (17, 18) to be connected selectively to either said second port means (19, 20) or to said third port means (21, 22).
5. A valve block according claim 4, **characterized in that** the reversing valve (2) is manoeuvred electro magnetically (3).
6. A tool attachment, **characterized in that** the attachment is provided with a valve block (1) according to any one of claims 1 - 5.
7. A working machine, **characterized in that** the machine includes a tool attachment according to claim 6.
8. The use of a valve block according to any one of claims 1 - 5 for connecting and manoeuvring a working tool or implement.

Patentansprüche

1. Ventilblock (1) mit einem Leitungssystem zur Verwendung bezogen auf ein Anbauwerkzeug einer Arbeitsmaschine, wobei das Anbauwerkzeug zur Verbindung mit unterschiedlichen Arten von hydraulisch betriebenen Werkzeugen oder Geräten mithilfe einer hydraulisch betätigten Verriegelungsvorrichtung (7) ausgelegt ist, dass der Ventilblock (1) erste Anschlussmittel (17, 18) zum Verbinden mit Hydraulikleitungen (32, 33) der Arbeitsmaschine, zweite Anschlussmittel (19, 20) zum Verbinden mit hydraulischen Verbindungsmitteln (10), die zum Verbinden mit hydraulischen Verbindungselementen (12) des Werkzeugs oder Geräts ausgelegt sind, und dritte Anschlussmittel (21, 22) zum Verbinden mit der Verriegelungsvorrichtung (7) aufweist, dass ein Ventilmittel (2) in dem Ventilblock (1) mit einer Funktion vorgesehen ist, die ermöglicht, dass die ersten Anschlussmittel (17, 18) gezielt mit entweder den zweiten Anschlussmitteln (19, 20) oder den dritten Anschlussmitteln (21, 22) verbunden werden, dass das

Leitungssystem mindestens eine Verbindungsleitung (40, 41) aufweist, die mit einem Ventilmittel in Form eines Rückschlagventils (6) versehen ist und die die zweiten Anschlussmittel (19, 20) und die dritten Anschlussmittel (21, 22) verbindet, damit ein Nachspannen der Verriegelungsvorrichtung (7) erreicht wird, wenn die ersten Anschlussmittel (17, 18) mit den zweiten Anschlussmitteln (19, 20) verbunden sind, und zum Entlasten vor dem Anschließen und Trennen der hydraulischen Verbindung (12) des Arbeitswerkzeugs; **dadurch gekennzeichnet, dass** das Leitungssystem im Ventilblock (1) mit den Anschlussmitteln in Verbindung steht und ebenfalls Ventilmittel (4, 5) aufweist, wobei die Ventilmittel (4, 5) so ausgelegt sind, dass sie mindestens eine der folgenden Funktionen erfüllen:

- eine erste Funktion zum Sicherstellen des Vorhandenseins von Flüssigkeitsdruck in der Verriegelungsvorrichtung (7) im Falle eines Bruchs oder einer Undichtigkeit in den hydraulischen Leitungen (32, 33) der Arbeitsmaschine durch ein erstes Rückschlagventil (4), das die erste Funktion erfüllen soll; und
- eine zweite Funktion zum Erreichen einer Druckbegrenzung der Verriegelungsvorrichtung (7), dadurch, dass die dritten Anschlussmittel (21, 22) zwei Anschlüsse (21, 22) aufweisen, die jeweils mit einer Hydraulikleitung (38, 39) verbunden sind, und dadurch, dass zwischen den Hydraulikleitungen eine Verbindungsleitung (42) vorgesehen ist, die ein Überdruckventil (5) aufweist und die so ausgelegt ist, dass sie die zweite Funktion erfüllt.

2. Ventilblock (1) nach Anspruch 1, **dadurch gekennzeichnet, dass** die Ventilmittel (4, 5) so ausgelegt sind, dass sie alle diese Funktionen erfüllen.
3. Ventilblock (1) nach Anspruch 1 oder 2, **dadurch gekennzeichnet, dass** der Ventilblock (1) Mittel (23) zum Befestigen des Ventilblocks (1) an der Verriegelungsvorrichtung (7) aufweist.
4. Ventilblock nach einem der Ansprüche 1 bis 3, **dadurch gekennzeichnet, dass** die Ventilmittel (2, 4, 5, 6) ein Umschaltventil (2) aufweisen, das so ausgelegt ist, dass es die Funktion erfüllt, die ermöglicht, dass die ersten Anschlussmittel (17, 18) gezielt mit entweder den zweiten Anschlussmitteln (19, 20) oder mit den dritten Anschlussmitteln (21, 22) verbunden werden.
5. Ventilblock nach Anspruch 4, **dadurch gekennzeichnet, dass** das Umschaltventil (2) elektromagnetisch (3) betätigt wird.
6. Anbauwerkzeug, **dadurch gekennzeichnet, dass**

das Anbaugerät mit einem Ventilblock (1) nach einem der Ansprüche 1 bis 5 versehen ist.

7. Arbeitsmaschine, **dadurch gekennzeichnet, dass** die Maschine ein Anbauwerkzeug nach Anspruch 6 aufweist.
8. Verwendung eines Ventilblocks nach einem der Ansprüche 1 bis 5 zum Verbinden und Betätigen eines Arbeitswerkzeugs oder Geräts.

Revendications

1. Bloc distributeur (1) comprenant un système de lignes à utiliser en relation avec un porte-outil d'une machine de travail, dans lequel le porte-outil est adapté pour être connecté à différents types d'outils ou d'instruments actionnés de façon hydraulique à l'aide d'un dispositif de verrouillage à commande hydraulique (7), que le bloc distributeur (1) comprend des premiers moyens de port (17, 18) à connecter à des lignes hydrauliques (32, 33) de la machine de travail, des deuxièmes moyens de port (19, 20) à connecter à des moyens de couplage hydrauliques (10) adaptés pour être couplés à des éléments de couplage hydrauliques (12) de l'outil ou de l'instrument, et des troisièmes moyens de port (21, 22) à connecter au dispositif de verrouillage (7), dans lequel des moyens de soupape (2) sont prévus dans le bloc distributeur (1) avec une fonction qui permet auxdits premiers moyens de port (17, 18) d'être connectés de façon sélective soit auxdits deuxièmes moyens de port (19, 20), soit auxdits troisièmes moyens de port (21, 22), dans lequel le système de lignes comprend au moins une ligne de connexion (40, 41) qui est pourvue de moyens de soupape se présentant sous la forme d'un clapet de retenue (6) et qui connectent lesdits deuxièmes moyens de port (19, 20) et lesdits troisièmes moyens de port (21, 22) de manière à obtenir un post-tensionnement du dispositif de verrouillage (7) lorsque les premiers moyens de port (17, 18) sont connectés auxdits deuxièmes moyens de port (19, 20) et pour relâcher une charge avant la connexion et la déconnexion de la connexion hydraulique (12) de l'outil de travail; **caractérisé en ce que** le système de lignes dans le bloc distributeur (1) communique avec lesdits moyens de port et comprend également des moyens de soupape (4, 5), dans lequel lesdits moyens de soupape (4, 5) sont adaptés pour remplir au moins une des fonctions suivantes:
 - une première fonction pour assurer la présence d'une pression hydraulique dans le dispositif de verrouillage (7) dans le cas d'une cassure ou d'une fuite dans les lignes hydrauliques (32, 33)

de la machine de travail par un premier clapet de retenue (4) qui est destiné à exécuter ladite première fonction; et
 - une deuxième fonction pour atteindre une limitation de pression du dispositif de verrouillage (7), 5

en ce que les troisièmes moyens de port (21, 22) comprennent deux ports (21, 22) qui sont chacun connectés à une ligne hydraulique (38, 39), et 10
en ce que, entre ces lignes hydrauliques, il est prévu une ligne de connexion (42) qui comprend une soupape de limitation de pression (5) et qui est adaptée pour exécuter ladite seconde fonction. 15

2. Bloc distributeur (1) selon la revendication 1, **caractérisé en ce que** lesdits moyens de soupape (4, 5) sont adaptés pour exécuter la totalité desdites fonctions. 20
3. Bloc distributeur (1) selon l'une quelconque des revendications 1 et 2, **caractérisé en ce que** le bloc distributeur (1) comprend des moyens (23) pour attacher le bloc distributeur (1) au dispositif de verrouillage (7). 25
4. Bloc distributeur selon l'une quelconque des revendications 1 à 3, **caractérisé en ce que** les moyens de soupape (2, 4, 5, 6) comprennent une soupape d'inversion (2) qui est adaptée pour exécuter ladite 30
 deuxième fonction qui permet auxdits premiers moyens de port (17, 18) d'être connectés de façon sélective soit auxdits deuxièmes moyens de port (19, 20), soit auxdits troisièmes moyens de port (21, 22). 35
5. Bloc distributeur selon la revendication 4, **caractérisé en ce que** la soupape d'inversion (2) est manoeuvrée de façon électromagnétique (3). 40
6. Porte-outil, **caractérisé en ce que** le porte-outil est pourvu d'un bloc distributeur (1) selon l'une quelconque des revendications 1 à 5. 45
7. Machine de travail, **caractérisée en ce que** la machine comprend un porte-outil selon la revendication 6. 50
8. Utilisation d'un bloc distributeur selon l'une quelconque des revendications 1 à 5 pour connecter et manoeuvrer un outil ou un instrument de travail. 55

Fig. 1

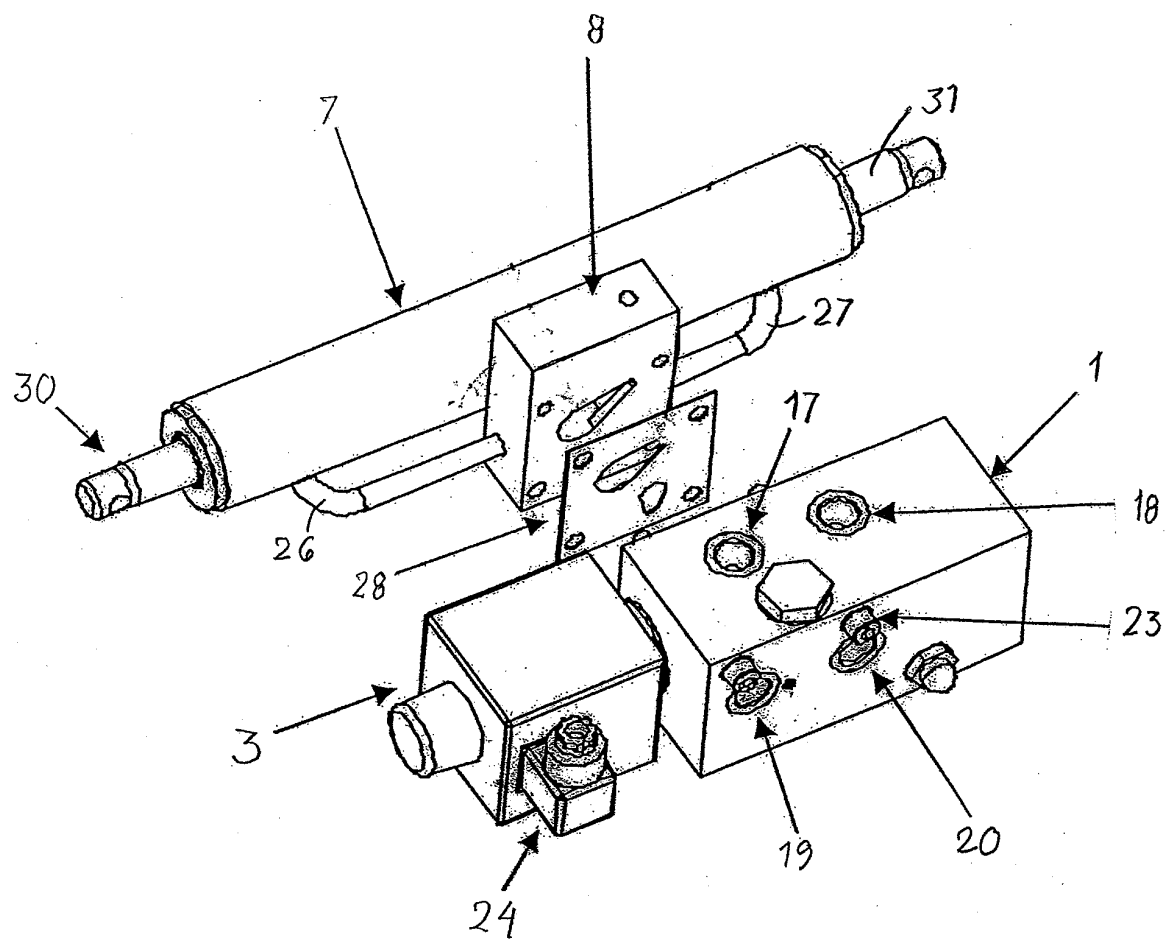
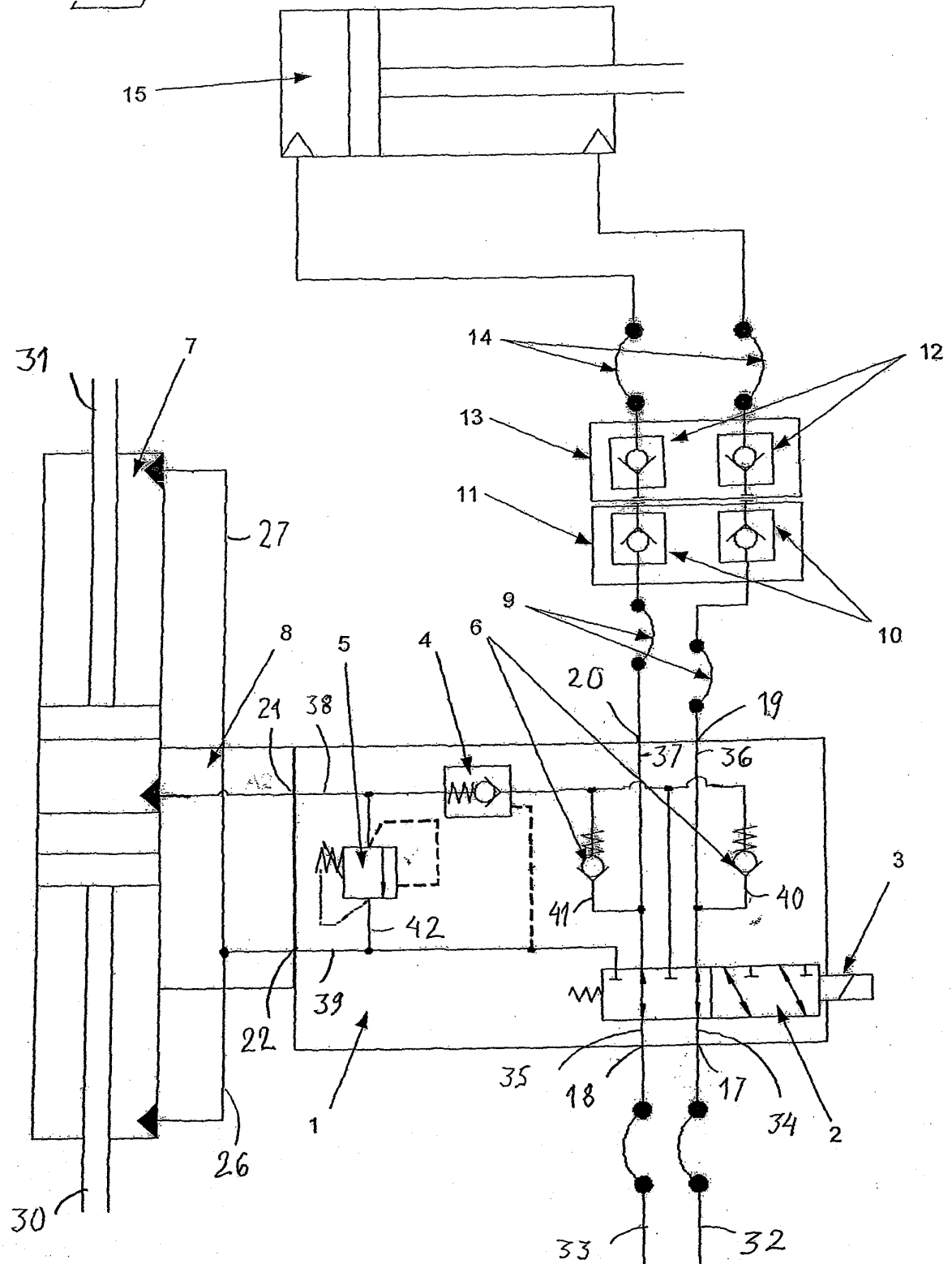


Fig. 2



REFERENCES CITED IN THE DESCRIPTION

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